Apply filters to SQL queries

Project description

My organization is focused on improving system security. As a security analyst, my responsibilities include monitoring potential security threats, investigating suspicious activity, and updating employee systems when necessary.

To complete these tasks, I used SQL queries with filters to retrieve specific information related to login attempts and employee machines.

Retrieve after hours failed login attempts

A potential security incident occurred after business hours, specifically after 18:00. All failed login attempts during this period needed investigation.

To identify these events, I created a SQL query that filtered for:

- Login attempts occurring after 18:00
- Unsuccessful login attempts

SQL Query

```
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00' AND success = FALSE;
```

Explanation

- `SELECT *` retrieves all columns from the `log_in_attempts` table.
- `login_time > '18:00'` filters attempts made after business hours.
- `success = FALSE` filters only failed login attempts.
- The `AND` operator ensures both conditions must be true.

This query helped identify suspicious failed access attempts occurring outside normal working hours.

Retrieve login attempts on specific dates

A suspicious event was detected on `2022-05-09`. Login attempts on that date and the previous day required investigation.

```
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09'
OR login_date = '2022-05-08';
```

Explanation

- The query retrieves all records from the `log_in_attempts` table.
- `login_date = '2022-05-09'` filters attempts on the suspicious date.
- `login_date = '2022-05-08'` filters attempts from the day before.
- The `OR` operator returns records matching either condition.

This allowed investigation of all login activity surrounding the suspicious event.

Retrieve login attempts outside of Mexico

After reviewing login records, I identified suspicious login attempts originating outside of Mexico.

SQL Query

```
SELECT *
FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

Explanation

- `NOT` excludes records matching the condition.
- `LIKE 'MEX%'` matches values beginning with `MEX`, including:
    - `MEX`

    - MEXICO
- % is a wildcard representing any number of characters.

This query returned all login attempts from countries other than Mexico.

Retrieve employees in Marketing

Employees in the Marketing department located in the East building required system updates.

SQL Query

```
SELECT *
FROM employees
WHERE department = 'Marketing'
AND office LIKE 'East%';
```

Explanation

- `department = 'Marketing'` filters Marketing employees.
- `Office LIKE 'East%'` filters offices located in the East building.
- The % wildcard allows matching office numbers such as:
    - `East-101`
    - `East-205`

The AND operator ensures employees meet both conditions.

Retrieve employees in Finance or Sales

Employees in the Finance and Sales departments required a separate security update.

SQL Query

```
SELECT *
FROM employees
WHERE department = 'Finance'
OR department = 'Sales';
```

Explanation

- `department = 'Finance'` filters Finance employees.
- `department = 'Sales'` filters Sales employees.
- The `OR` operator returns employees from either department.

This query helped identify all systems requiring the additional update.

Retrieve all employees not in IT

Another update needed to be applied to employees outside the Information Technology department.

SQL Query

```
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

Explanation

- `NOT` excludes employees belonging to the Information Technology department.
- All remaining employee records are returned.

This allowed the team to identify systems needing the final security update.

Summary

In this project, I used SQL filtering techniques to retrieve specific cybersecurity-related information from the `log_in_attempts` and `employees` tables.

The queries involved:

- Investigating suspicious login activity
- Identifying failed login attempts
- Filtering login events by date and country
- Retrieving employee systems requiring security updates